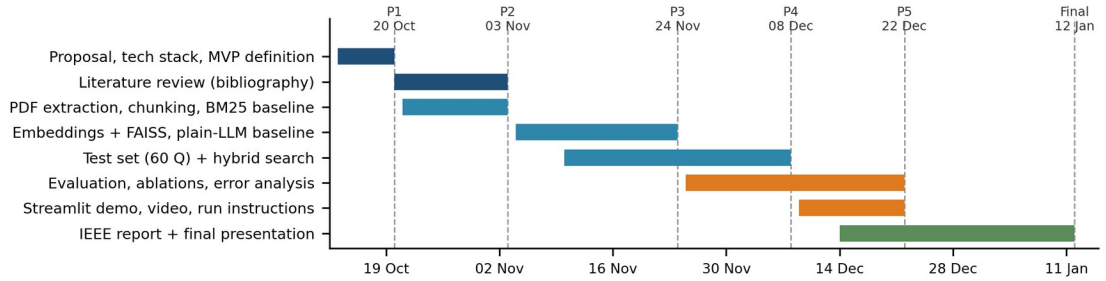
MDM2 – Case Study: Intelligent Systems in Production
One-Page Proposal

Team	FactoryDocs Assistant team
Members	**Divi Teja Dimmiti** (Team Lead – RAG pipeline & LLM integration) · **Shilpa Golla** (PDF extraction & LLM integration) · **Vijay Boyapati** (evaluation & Streamlit demo); test set shared by all
Project Title	FactoryDocs Assistant: A Retrieval-Augmented Chatbot for Robot Error Diagnosis in Production
GitHub Repository URL	https://github.com/Shilpa-Golla/FactoryDocs-Assistant-A-Retrieval-Augmented-Chatbot-for-Production-and-Logistics-Manuals
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Industrial Application (target domain/use-case)	Robot error diagnosis in production: when a UR5e cobot (e.g., loading a CNC machine) triggers a protective stop, the operator types the error code or describes the problem and gets the cause, fix steps and source pages from the robot and gripper manuals.
Keywords (3–6)	Retrieval-Augmented Generation; Robot Error Diagnosis; Collaborative Robots; Hybrid Search; Local LLM; Downtime Reduction
Submission Date (YYYY-MM-DD)	2026-10-20

Gantt Chart (P1–P5 = course phases)



1) Problem Statement & Measurable Outcomes (3–4 sentences)	**Problem:** when a UR5e cobot triggers a protective stop (e.g., C153 position deviates from path, C157 collision detected by joint), the connected machine and material flow stop; UR requires the root cause to be reviewed before reset. Causes such as wrong payload, collisions or dragging cables, and their fixes, are spread over four long manuals; workers describe problems in their own words, which Ctrl+F cannot match, and general chatbots give unverified answers. We build a local chatbot that answers only from these manuals, with source pages. **KPIs (60-question test set):** (1) retrieval hit rate@5 ≥ 80%; (2) answer accuracy ≥ 75%, beating BM25 and plain-LLM baselines; (3) correct "I don't know" on ≥ 80% of unanswerable questions; (4) median response time ≤ 10 s on a laptop.
2) Motivation & Industrial Relevance (2–3 sentences)	Unplanned downtime costs the world's 500 largest companies about $1.4 trillion per year (Siemens, 2024); faster root-cause finding shortens every robot stop. Operators and technicians benefit most when no expert is nearby; a local model keeps documents on site.
3) Related Work Snapshot (2–3 key references)	[1] P. Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," NeurIPS 2020. [2] Y. Gao et al., "Retrieval-Augmented Generation for Large Language Models: A Survey," arXiv:2312.10997, 2023. [3] S. Es et al., "RAGAS: Automated Evaluation of Retrieval Augmented Generation," EACL 2024 (Demos). **Gap:** RAG is mostly evaluated on general text; few studies address robot error diagnosis across several manuals, with queries mixing exact error codes and free-text problems, or measure refusal with a local model.
4) Method & Feasibility (≤6 sentences)	**Data:** four public manuals – UR Error Codes Directory, UR5e User Manual, Robotiq 2F-85/2F-140 gripper manual (for UR), UR e-Series Service Handbook (linked, not redistributed). **Pipeline:** pdfplumber extracts text and tables → page-tagged chunks → sentence-transformers embeddings in FAISS + BM25 index → local LLM (Ollama, e.g., Llama 3 8B) gives meaning, cause and fix with source pages, or refuses. **Comparison:** plain LLM vs. BM25 RAG vs. dense RAG vs. hybrid RAG, plus chunk-size and top-k tests. **Evaluation:** 60 team-written questions in 5 types (error code, problem description, multi-document, procedure, unanswerable), scored by two members (3-level rubric) and RAGAS. **Feasibility & artifacts:** free tools, laptop/Colab (≈ €0); GitHub repo, Streamlit demo, video, IEEE report.
5) Milestones & Timeline (short table/list)	**P1 20 Oct:** proposal, tech stack, MVP (revised after feedback) · **P2 03 Nov:** literature review, PDF extraction, BM25 baseline, test plan **P3 24 Nov:** embeddings + FAISS, first cited answers, plain-LLM baseline · **P4 08 Dec:** test set, hybrid search, comparison + error analysis **P5 22 Dec:** Streamlit demo, demo video, run instructions · **Final 12 Jan 2027:** IEEE report and presentation
6) Risks & Ethics (1–2 sentences)	Decision aid only: UR requires trained personnel to review every protective stop, and the assistant supports this, never replaces it; manuals linked, no personal data. Risks: broken PDF table extraction (compare pdfplumber vs. PyMuPDF) and slow local inference (smaller quantized model).

Phase 1 rubric (15%): Team & GitHub 2% · On-time 2% · Topic & Proposal 6% · Presentation 5%